Timing is everything: EEG and Predictive Coding simulations show that reading conditions influence whether late frontal positivities track misprediction or lexical predictability

Introduction. Decades of research show the N400 (300–500ms) ERP component is reliably modulated by contextual predictability.[1] During deep comprehension, unexpected but plausible words that disconfirm strong predictions in high-constraint contexts also evoke a later frontal positivity (LFP, 500–1000ms).[2,3] These "misprediction" LFP effects are argued to reflect the detection and resolution of residual misalignment when higher-level states continue to generate outdated top-down predictions. However, LFP effects appear to vary considerably across studies: In RSVP studies with young adults, they are selectively larger to mispredicted inputs,[2,3] whereas in studies with slower presentation rates, self-paced reading, or older adults, LFPs can occur in low-constraint contexts and seem to be graded by contextual predictability (similar to the N400).[4,5] Here, we ask two questions: First, during self-paced reading of short vignettes, are LFPs routinely evoked by each content word, scaling with their predictability, regardless of whether the context predicts a more likely alternative? Second, can a single predictive coding (PC) framework (Figure 1) explain why "misprediction" and "graded" LFPs arise under different reading conditions?

Study 1 (ERPs): We recorded EEG while 22 native English-speakers self-paced through short vignettes. We measured the N400 (300–500ms) and LFP (500–1000ms) for each content word. Predictability values came from an online cloze task (N=361). Mixed effects models revealed effects of Predictability (target probability) but not Misprediction (probability of the most common alternative) on both components, with smaller responses for more predictable words (Figure 2).

Study 2 (Simulations): We used a PC model[6] to simulate different reading conditions by varying the timing and overlap of top-down predictive pre-activation of the highest conceptual level, and bottom-up activation of the lowest orthographic level (Figure 3). We defined the N400 and LFP as the magnitudes of bottom-up prediction error (buPE) and top-down bias (tdB), respectively, at each iteration until the model converged. In both simulations, the buPE (N400) was only sensitive to target pre-activation. In contrast, the effects of pre-activation and misprediction on the tdB (LFP) depended on when pre-activation began and whether it overlapped with bottom-up activation. If top-down pre-activation started slightly before and continued during bottom-up processing, the tdB resembled the "misprediction" LFPs in prior RSVP studies (selectively larger to unexpected but plausible inputs). In contrast, if top-down pre-activation began well before bottom-up activation, the tdB resembled LFPs in our self-paced study, with its magnitude decreasing with increasing predictability. In this case, the tdB was not driven by mispredictions. Instead, it reflected sustained residual activation of irrelevant semantic features from orthographic neighbors co-activated during bottom-up processing, and it was attenuated for expected inputs because strong top-down predictions suppressed this irrelevant activation.

Discussion. These data suggest that LFPs can be driven by multiple sources of misalignment, whose relative contributions trade off under various reading conditions. Selective LFPs to mispredicted words are more likely during RSVP because outdated top-down predictions continue to conflict with the bottom-up input. In contrast, less selective LFPs that scale with predictability are more likely during self-paced reading because longer bottom-up processing allows irrelevant residual activation to persist for unexpected words, while earlier top-down lexical predictions suppress this residual activation for expected words.

References: [1] Kutas & Federmeier, 2011; [2] Federmeier, 2007; [3] Kuperberg et al., 2020; [4] Payne & Federmeier, 2017; [5] Wlotko & Federmeier, 2012; [6] Nour Eddine, 2024.

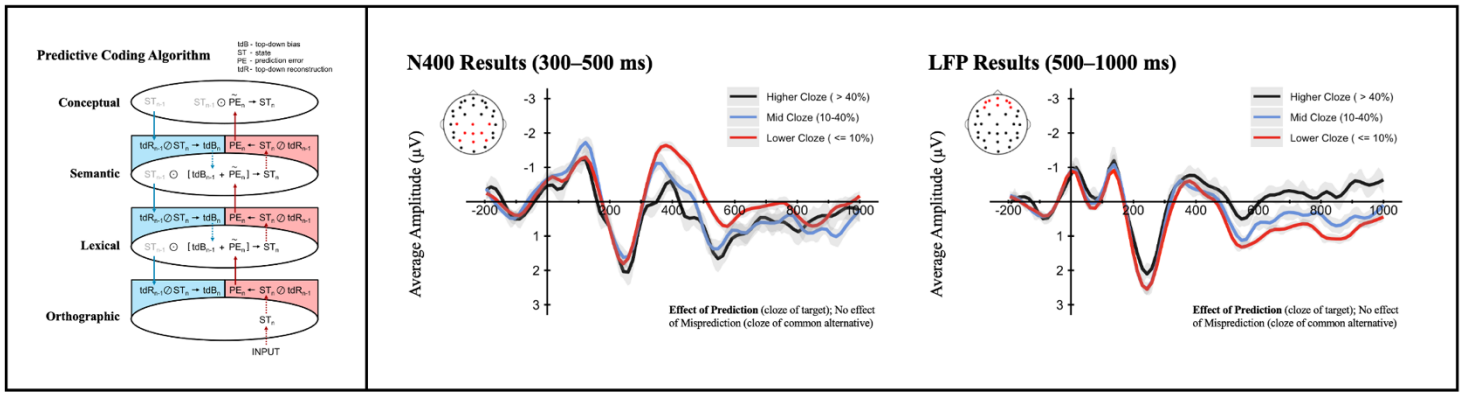

Figure 1. *Architecture and algorithm of Predictive Coding Model (Right panel).*
Figure 2. *Self-paced reading ERP results (Left panel).* Grand-average ERPs at centroparietal (N400; left) and frontal (LFP; right) sites, binned by target predictability (cloze). Trial-level linear mixed models showed main effects of target cloze on N400 and LFP amplitudes, but no effect of the probability of the most common alternative continuation. Models included z-scored nuisance covariates (ONSize, frequency, concreteness).

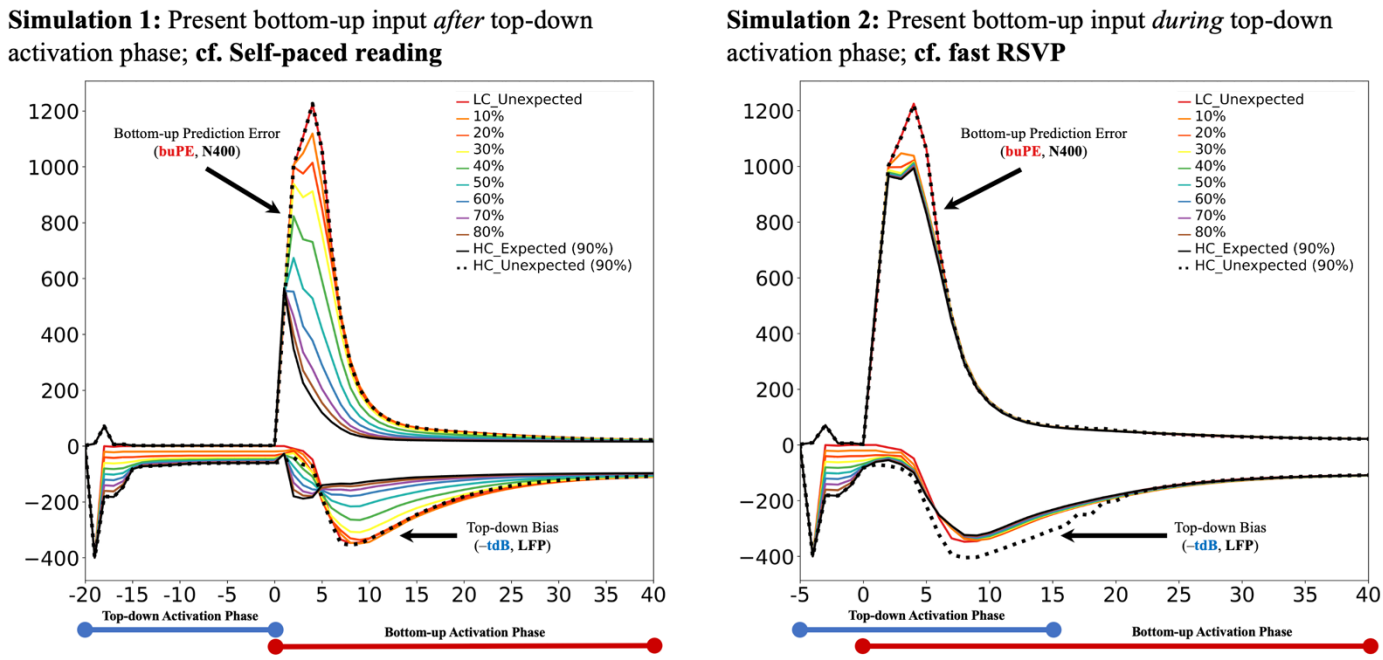

Figure 3. *Predictive coding simulation results.* (Left; Simulation 1) The expected target was pre-activated for 20 iterations (0–90%), beginning 20 iterations before bottom-up presentation of an expected or unexpected word (presented for 20 iterations). (Right; Simulation 2) Same, but pre-activation began 5 iterations before bottom-up presentation. Iterations (x-axis) and arbitrary model activation units (y-axis).